LUONG TIEN DUNG

SOFTWARE ENGINEERING / ALGORITHM INTERN

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GitHub: github.com/Luong-dung | LeetCode: leetcode.com/u/__covetot__

Third-year Data Science student focused on C++ algorithm design and optimization. Designed a macro-based N-Puzzle strategy and benchmarked large-scale test cases.

EDUCATION

VNU University of Science (HUS), Vietnam National University, Hanoi
Bachelor's in Data Science | Expected 2028 | GPA: **3.11/4.00**

TECHNICAL SKILLS

Programming: C++, Python, Java, JavaScript, SQL
Algorithms: Data structures, graph algorithms, dynamic programming, search, optimization
Tools: Git, GitHub, Linux, Jupyter Notebook

SELECTED PROJECTS

Large-Scale N-Puzzle Solver

University group project | C++ | Algorithm design & benchmarking

- Designed the **core macro-based strategy**, decomposing the puzzle through alternating row and column operations; used **BFS fallback** for exceptional configurations.
- Compared BFS, IDA*, and hybrid search approaches through iterative experiments to assess runtime, memory use, and scalability.
- Benchmarked an **AI-optimized solver** on **1000 × 1000 (1 million tiles)** boards: **21/30 solved (70%)** within **10 seconds / 1 GB RAM**; successful runs averaged **9.00 s**.
- Directed **AI-assisted implementation** of the original strategy and iterative optimization based on benchmark results.

Sign Language Recognition (Video Classification)

Independent experiments | Python, PyTorch, OpenCV

- Built a **100-class video classification** pipeline covering frame sampling, preprocessing, augmentation, training, and inference.
- Tested pretrained CNNs and Transformer-based temporal modeling; investigated data leakage and class imbalance. Achieved **74.57% Macro-F1 (0.7457) on the private test set** of the **2025 OLP AI task** used for practice.

Local Coding Agent

Personal project | Python, Ollama

- Connected local LLM-generated code to **C++ compilation, execution, and automated tests**; used test feedback in iterative code-improvement experiments.

COMPETITIVE PROGRAMMING & AWARDS

Third Prize (2025) — Vietnam Student Informatics Olympiad, Specialized Division
Honorable Mention (2024) — Vietnam Student Informatics Olympiad, Specialized Division
LeetCode: Contest Rating **1807** · **280** problems solved | **AtCoder:** Peak Rating **1610**

Available part-time during the academic year; open to full-time work in summer 2027.